



Department of Computer Science and Engineering

Jashore University of Science and Technology

4th Year 1st Semester Final Examination-2025 (Session: 2021-22)

Course Code: BUS 4101

Course Title: E-commerce and Management Information System

Time: 3.00 hours

Marks: 72

(Note: Answer any six set of questions from the following eight set of questions)

1. a) Briefly explain the three-tier E-Commerce architecture. 03
b) Differentiate between E-commerce and F-commerce. 04
c) Describe e-commerce business models with suitable examples. 05
2. a) A startup company plans to create an online platform where individual users can sell used products directly to other consumers. The company earns revenue through transaction fees and advertisements. 3×2
i) Identify the appropriate e-commerce business model.
ii) Explain how the model operates.
iii) Discuss two advantages and two challenges of this model.
b) Describe the Business-to-Business (B2B) model with two examples. What are its advantages and limitations? 06
3. a) List three measures to prevent cyber fraud in Mobile Financial Services. 03
b) Using the RFM model, classify the following customers as high-value or low-value and suggest one suitable marketing strategy for each. 04
i) Customer A: Purchased 2 weeks ago, made 8 purchases this year, and spent BDT 12,000.
ii) Customer B: Purchased 6 months ago, made 1 purchase, and spent BDT 1,000.
- c) Consider the following activity table for a project. 05

Activity	Predecessors	Duration (Weeks)
P	-	3
Q	P	3
R	-	2
S	Q,R	1
T	R,S	2
U	T	3
V	S,U	1

Draw a network diagram and apply CPM to find the critical path for the project.

4. a) Differentiate between the Drop-shipping and In-house fulfillment models. 03
b) Using Anthony's Triangle, explain how TPS, MIS, DSS, and EIS support different management levels. 04
c) An e-commerce electronics seller sells 12,000 units/year of a specific power bank. Each purchase order costs 500 BDT to process. Holding one unit in the warehouse costs 50 BDT/year. Calculate the EOQ and the optimal number of orders per year. 05
5. a) State Metcalfe's Law and explain its importance in network economics. 03
b) A platform grows from 100,000 to 200,000 users. By what factor does the network value increase? 04
c) A company is evaluating two investment strategies: 05
Strategy A: 60% chance of BDT 8,00,000 profit and 40% chance of BDT 2,00,000 loss.
Strategy B: 45% chance of BDT 10,00,000 profit and 55% chance of BDT 1,00,000 loss.
Calculate the Expected Value (EV) of each strategy and recommend the better strategy based on the expected value.

6. a) Define each level of the DIKW hierarchy with suitable examples. 03
- b) Why is it difficult for a new company to compete with platforms such as Google, Facebook, or Daraz? Explain using the concepts of network effects and winner-takes-all dynamics. 04
- c) A company implemented an ERP/MIS; 65% of shop-floor employees resisted. Using Kotter's Change Model, list the steps and describe the specific action management should take at the most critical steps to address resistance. 05
7. a) Discuss the common security threats faced by e-commerce websites and client-server systems. 10
- b) Explain the role of encryption, digital signatures, and SSL/TLS in securing e-commerce transactions. 02
8. a) Define the CIA Triad. 03
- b) bKash holds transaction history of 70 million users. The BFIU proposes sharing anonymized data with government agencies for financial crime detection. Apply Mason's PAPA framework to identify ethical concerns for each dimension. 04
- c) A bank's EIS alerts: "Branch X generates 58% of losses with 18% of customers." Describe, step by step, how the CEO uses the EIS drill-down feature to investigate the problem and recommend one strategic decision. 05



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Semester Final Examination-2026
4th Year 1st Semester, Session: 2021-22

Course Code: CSE 4105
Course Title: Computer Security

Time: 3 hours
Full Marks: 72

(N.B. answer any 6 set questions from the following 8 set questions. The figures shown in the margin indicate full marks.)

1.
 - a) Define cyber security. Explain why cyber security is essential in modern organizations. (3)
 - b) Describe security goals. Explain the CIA triad with suitable real-life examples. (4)
 - c) Define trojan horse attack. Explain active attacks and passive attacks with proper examples. (5)
2.
 - a) Define cryptography. Differentiate between substitution cipher and transposition cipher. (4)
 - b) What is Botnet? Explain the model for network security with proper diagram. (4)
 - c) Explain transposition cipher. Encrypt the plaintext "COMPUTERNETWORKS" using a 3-rail rail fence cipher, then decrypt the resulting ciphertext to recover the original message. (4)
3.
 - a) Define RSA algorithm. Mention the disadvantages of playfair cipher. (3)
 - b) Explain steganography. Compare it with cryptography and mention three practical applications. (3)
 - c) Mention the steps of the RSA algorithm. Using $p = 11$ and $q = 13$, generate a valid pair of RSA public and private keys by selecting a suitable public exponent e . Encrypt the message $M = 9$, then decrypt the ciphertext to verify the original message. (6)
4.
 - a) Differentiate between symmetric key cryptography and asymmetric key cryptography. (3)
 - b) Explain playfair ciphers with proper example. (6)
 - c) Describe incident management. Show the network vulnerability assessment method. (3)
5.
 - a) What is the main difference between vigenère cipher and vernam cipher. (2)
 - b) Explain the vernam cipher with proper example. (4)
 - c) What are the key information kept in a digital certificate? Explain the role of Certification Authority (CA) with necessary figure. (6)
6.
 - a) Define trap door and logic bomb with suitable example. (3)
 - b) Explain columnar transposition cipher with proper example. (3)
 - c) Differentiate between virus and worm. (2)
 - d) Explain the working principle of a Distributed Denial-of-Service (DDoS) attack using a diagram. Also mention three effective mitigation techniques. (4)
7.
 - a) Define Doxing. Why is penetration testing important for an organization? (3)
 - b) Explain penetration testing. Mention the types of penetration testing with examples. (3)
 - c) What is a zero-day attack? Describe the phases of a standard penetration testing process. (4)
 - d) Differentiate between active reconnaissance and passive reconnaissance. (2)
8.
 - a) Explain the philosophy behind Diffie-Hellman key exchange protocol. In the Diffie-Hellman key exchange protocol, let the prime number be 353 and one of its prime root is 3. Let the users A and B select their secret keys 97 and 233, respectively. Compute the public keys of A and B and also their common secret key. (7)
 - b) Demonstrate all possible usages of a hash function for authentication, confidentiality and digital signature. (5)

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4th Year 1st Semester (Session: 2021-22)

Course Code: CSE-4115

Course Title: Software Design Patterns and Principles

Total Time: 3.00 hours

Total Marks: 72.0

(Note: Answer any six set of questions from the following eight set of questions)

- | | | |
|---|--|---|
| 1 | (a) Define software design patterns. Why are they important in software engineering? | 6 |
| | (b) What are the categories of design patterns? What are their purposes? Name some common design patterns under each category (at least two for each category). | 6 |
| 2 | (a) Differentiate between Factory pattern and Abstract Factory pattern using Class Diagram. In which way are they different? | 8 |
| | (b) What is the benefit of using Singleton pattern. Explain using code. | 4 |
| 3 | (a) A university is developing a Digital Library Management System. The system stores books, journals, magazines, and research papers collected from different digital repositories. Each repository provides metadata in different formats. The system must satisfy the following requirements: <ul style="list-style-type: none">• There should be only one Library Catalog throughout the application.• Different types of books (Printed Book, E-Book, Reference Book) should be created without exposing object creation logic.• Metadata received from external repositories should be converted into a common format.• All library resources should be processed uniformly so that the client can display them using the same iteration mechanism. | 9 |
| | (b) Draw the Class Diagram for the system described above. | 3 |
| 4 | (a) If there are many optional and mandatory attributes (name, date of birth, address, designation, department, research papers, etc.) in a class like Teacher, using which design pattern can we enable the object creation simple and readable? Explain with code. | 9 |
| | (b) Differentiate between a bug and a code smell. Name some common code smells (at least 18). | 3 |
| 5 | (a) Name the prevalent code smell in the following code: | 9 |

```
public class clsSalaryCalculator
{
    private double nBasicSalary = 0.0;
    private double nBonuspercent = 0.0;

    private AppointmentType eAppointmentType;

    public enum AppointmentType
    {
        Permanent,
        Temporary,
        Casual
    }

    public clsSalaryCalculator(double nBSalary, AppointmentType eAppType)
    {
        this.nBasicSalary = nBSalary;
        this.eAppointmentType = eAppType;
    }

    public double getBonus()
    {
        double nTotalSalary = 0.0;
```

```

        if (eAppointmentType == AppointmentType.Permanent)
        {
            nTotalSalary = nBasicSalary + nBasicSalary * 1.0;
            return nTotalSalary;
        }
        else if (eAppointmentType == AppointmentType.Temporary)
        {
            nTotalSalary = nBasicSalary + nBasicSalary * 0.5;
            return nTotalSalary;
        }
        else if (eAppointmentType == AppointmentType.Casual)
        {
            nTotalSalary = nBasicSalary + nBasicSalary * 0.1;
            return nTotalSalary;
        }
        else
        {
            throw new Exception("No Salary is defined.");
        }
    }
}

```

Using which design pattern can we eliminate the code smell? How?

- (b) Are Comments code smells? How can we eliminate that? Explain. 3
- 6 (a) A university is developing an Employee Hierarchy Management System to manage teachers, officers, and staff. The university wants to model its organizational structure so that a senior employee (such as a department head) can supervise multiple subordinate employees. 9
- The system should satisfy the following requirements:
- Every employee has personal information, designation, grade, appointment type, and salary.
 - Some employees, such as department heads can supervise one or more employees.
 - Subordinates may be teachers, officers, or staff.
 - The system should calculate the total salary expenditure of all subordinate employees (if any) under each employee.
 - The client application should treat an individual employee and subordinates in the same way without writing separate code for each case.
 - All employee types (Teacher, Officer, and Staff) should be handled uniformly so that their information can be displayed using the same loop.
- Identify and implement design patterns in the aforementioned Case Study.
- (b) Describe the Long Method code smell. How is it related to regression testing? 3
- 7 (a) Explain the Decorator design pattern with code. 6
- (b) Explain the Observer Design Pattern with a suitable real-world example. Implement the pattern using Java.. 6
- 8 (a) Explain the Open-Closed Principle (OCP) with a Class Diagram and Java implementation. 6
- (b) Describe the Data Clumps code smell. Explain appropriate refactoring techniques. 3
- (c) Explain the Middle Man code smell. How can it be removed through refactoring? 3



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Semester Final Examination-2025

4th Year 1st Semester, Session: 2021-2022

Course Code: CSE 4101

Time: 3.00 hours

Course Title: Computer Graphics and Multimedia

Marks: 72

[N.B. Answer any six set of questions from following eight set of questions. The figures in the right margin indicate full marks.]

1. a) Explain the graphical applications used in building artificial aircraft. 3
b) What graphics methods are used in both fine art and commercial art applications? 2
c) What are the uses of graphical methods in computer science education? 3
d) Determine the next five points from the initial point $(x_0, y_0) = (0, b)$ for the first quadrant where radius $a=20$ and $b=10$. Use Table 1 as a guideline to find the next five points. 4

K	P_k	(x_{k+1}, y_{k+1})
0		
1		
2		
3		
4		
5		

2. a) Write the differences between Raster Scan Display and Random Scan Display. 3
b) Explain the methods for color display with a CRT. 3
c) Illustrate the architecture of a random-scan system with a display processor. 3
d) Briefly explain DDA algorithm. 3
3. a) Derive successive decision parameters values and positions along the circle path using the midpoint method. [Where $r=10$ and first quadrant from $x=0$ to $x=y$]. 4
b) Explain the steps of midpoint circle drawing algorithm. 4
c) Derive successive decision parameters for the midpoint ellipse algorithm assuming the start position is $(r_x, 0)$ and points are to be generated along the curve path in counterclockwise order. 4
4. a) Explain boundary fill algorithm with proper picture. 4
b) What is transformation? Explain translation and rotation in two-dimensional transformation. 3
c) Explain with example: 5
i) How to convert standard 3D coordinates, $(x; y; z)$, to homogeneous coordinates.
ii) How to convert homogeneous coordinates to standard 3D coordinates.
5. a) Define shades, tints, and tones. 3
b) With proper figure, explain general fixed point scaling. 3
c) With proper figure, explain two dimensional viewing pipeline. 4
d) Define- i) Depth Cueing ii) Surface Rendering 2
6. a) Describe the parametric continuity conditions of a spline. 3
b) Drive and explain the properties of B-Spline curves. 3
c) What is 3D object representation? List and define its categories. 3
d) What is perspective projection matrix? Derive the equation for it. 3
7. a) With proper figure explain general parallel projection transformation. 4
b) Briefly explain the operations of clipping. 5
c) How you can clip a text. Explain with an example. 3
8. a) Describe the properties of Bezier Curves. 4
b) Explain the steps for depth-buffer algorithm. 4
c) Briefly explain the basic functions of depth-sorting method. 4